Type MS · How many patterns from n bits

Answer 1 of 6 · Question 2 part d.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part d.i in this mark scheme extract

2(a)(i)	Award up to **two** marks for: • 0100 (1) 1110 (1)	Accept any number of leading zeros.	2
2(a)(ii)	**The only correct answer is B** *A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits* *C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits* *D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits*		1
2(a)(iii)	Award up to **two** marks for a linked explanation such as: All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1) Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1) Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1)	Accept the reverse argument Allow examples of non-latin characters	2

Question Number	Answer	Additional Guidance	Mark
2(b)	1010 1011 Award up to **two** marks for: • MSB = 1 (1) • Rest of pattern correct 010 1011 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(c)	Award up to **two** marks for: • 0100 (1) 1010 (1)		2

Type MS · How many patterns from n bits

Answer 1 of 6 · Question 2 part d.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part d.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:

• 3 bits used for all patterns (1)/
• No pattern repeated (1)

Example:

<table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:

• 3579 x 6128 x 32 (1)
• ÷ 8 (1)
• + 732 (1)
• ÷ (1000 x 1000) (1)

Examples

$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$

$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$

$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required
• Equivalent expressions are accepted
• Calculations not explicit but expressed gain the mark
• Award 1 mark for correct calculated answer of 87.73 if no other marks awarded</td><td>4</td></tr>
<tr><td></td><td></td><td>Total for question 2</td><td>15</td></tr>
</table>

Type MS · How many patterns from n bits

Answer 1 of 6 · Question 2 part d.i · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · How many patterns from n bits

Answer 2 of 6 · Question 4 part a.v · 2 marks

2020 NOV · Paper 01 · extract 1 of 4 · find part a.v in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · How many patterns from n bits

Answer 2 of 6 · Question 4 part a.v · 2 marks

2020 NOV · Paper 01 · extract 2 of 4 · find part a.v in this mark scheme extract

4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

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Type MS · How many patterns from n bits

Answer 2 of 6 · Question 4 part a.v · 2 marks

2020 NOV · Paper 01 · extract 3 of 4 · find part a.v in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

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Type MS · How many patterns from n bits

Answer 2 of 6 · Question 4 part a.v · 2 marks

2020 NOV · Paper 01 · extract 4 of 4 · find part a.v in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · How many patterns from n bits

Answer 3 of 6 · Question 1 part c · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · How many patterns from n bits

Answer 3 of 6 · Question 1 part c · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · How many patterns from n bits

Answer 4 of 6 · Question 1 part g.iii · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part g.iii in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · How many patterns from n bits

Answer 4 of 6 · Question 1 part g.iii · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part g.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · How many patterns from n bits

Answer 5 of 6 · Question 2 part b · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

Type MS · How many patterns from n bits

Answer 5 of 6 · Question 2 part b · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

Type MS · How many patterns from n bits

Answer 5 of 6 · Question 2 part b · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

Type MS · How many patterns from n bits

Answer 5 of 6 · Question 2 part b · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

Type MS · How many patterns from n bits

Answer 5 of 6 · Question 2 part b · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · How many patterns from n bits

Answer 6 of 6 · Question 3 part a.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

Type MS · How many patterns from n bits

Answer 6 of 6 · Question 3 part a.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

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Type MS · How many patterns from n bits

Answer 6 of 6 · Question 3 part a.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · How many patterns from n bits

Answer 6 of 6 · Question 3 part a.i · 1 mark

2024 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark